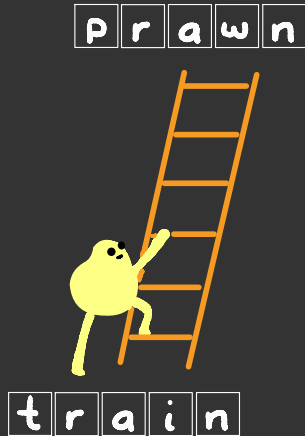


multiplayer games

CSCI
270

all the search tasks we've discussed
so far have involved a single agent

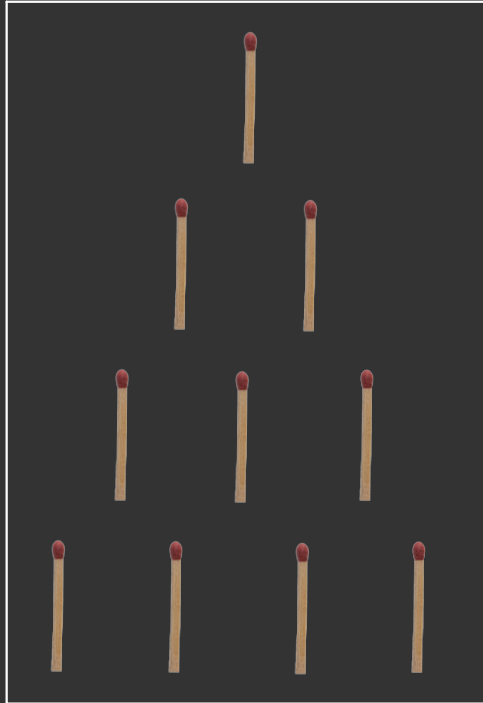


but what if others want to stop us from succeeding?



nim

on your turn,
you must remove
one or more
matchsticks
from a single row

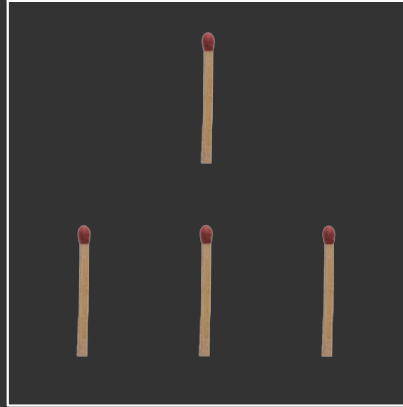


you lose if
you remove
the last
matchstick



nim

assume this initial configuration



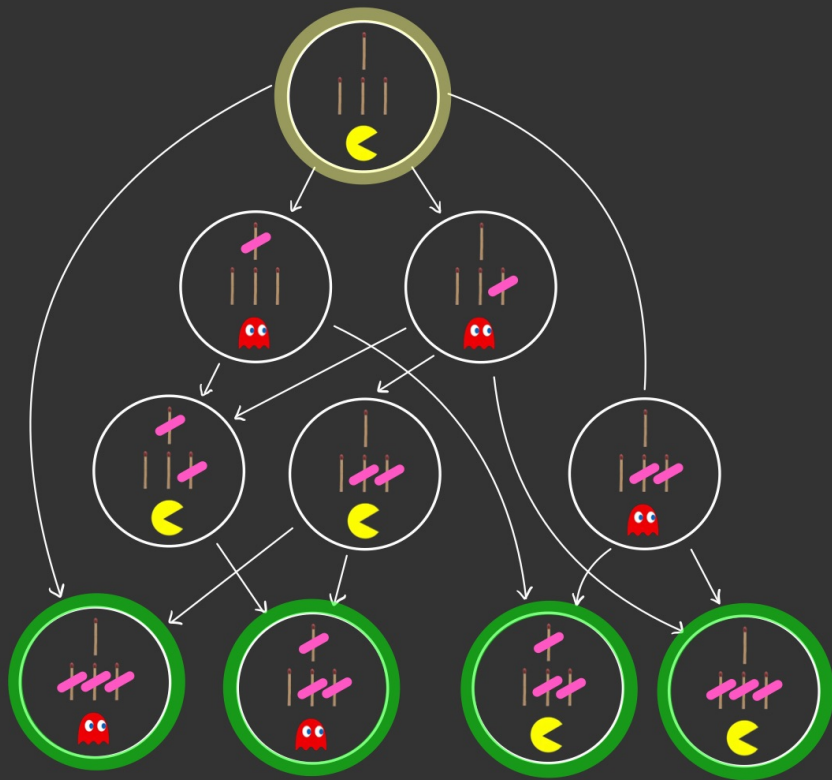
a **game** with players **P**
is a tuple $(\mathbf{m}, \mathbf{U})$ where:

► $\mathbf{m} = (\mathbf{Q}, \Sigma, \Delta, q_0, \mathbf{F})$

is a state machine where
each state $q \in \mathbf{Q}$ has the
form $(\mathbf{p}, q') \in \mathbf{P} \times \mathbf{Q}'$ for some
auxiliary set $\mathbf{Q}'$ of states

► $\mathbf{U} : \mathbf{F} \times \mathbf{P} \rightarrow \mathbb{R}$

is a function (called the
utility function) that
specifies the value of each
final state for each player



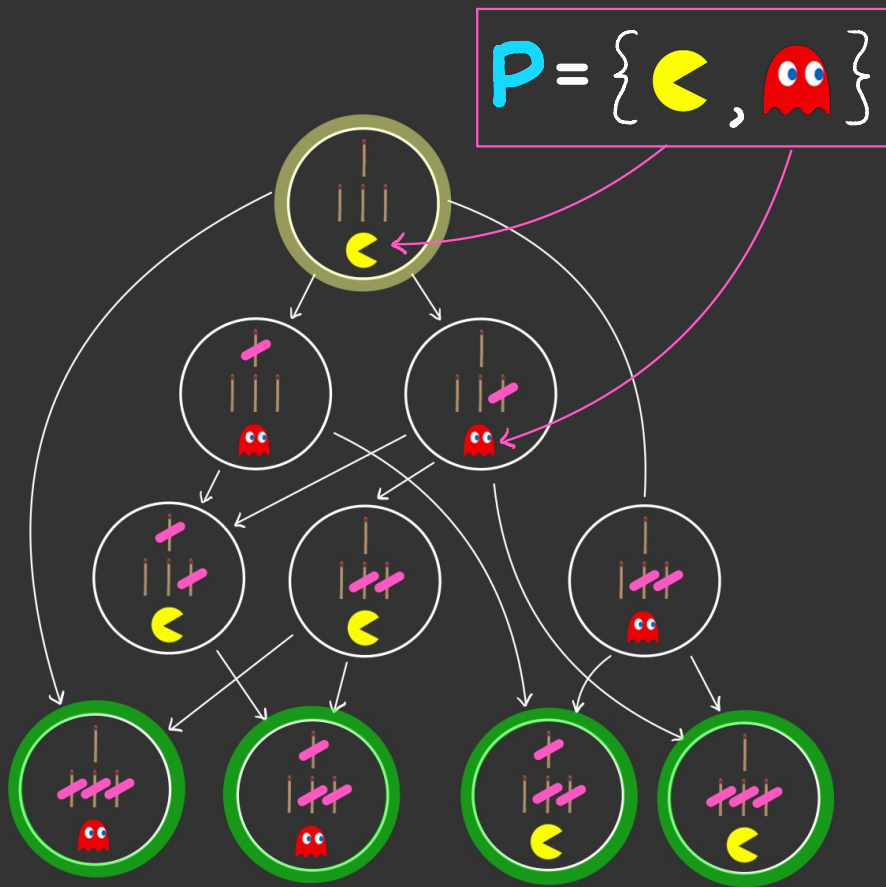
a game with players P
is a tuple (m, U) where:

► $m = (Q, \Sigma, \Delta, q_0, F)$

is a state machine where
each state $q \in Q$ has the
form $(p, q') \in P \times Q'$ for some
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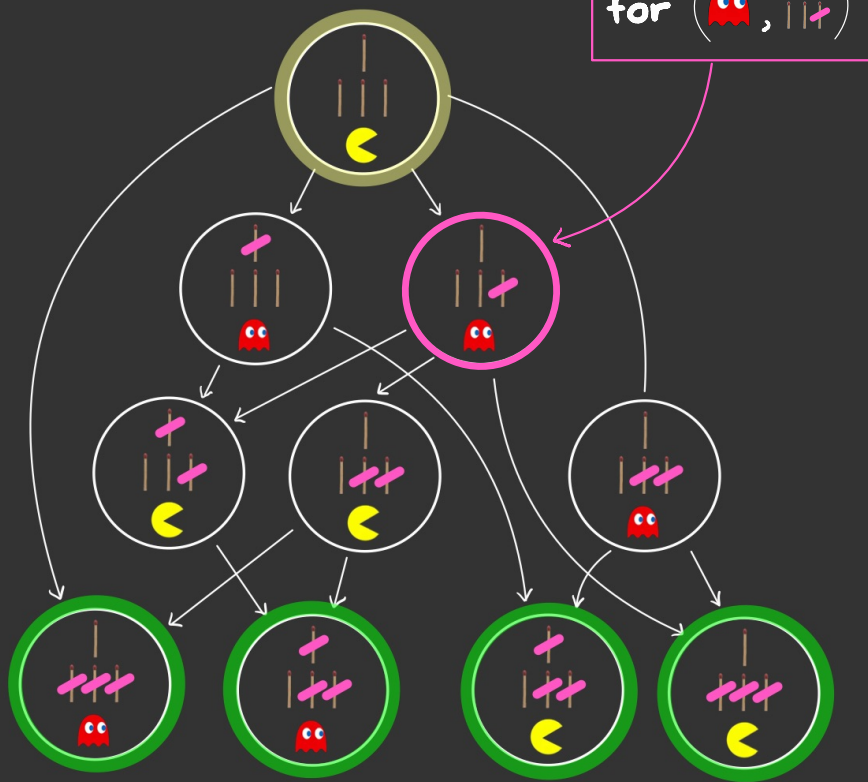
a game with players P
is a tuple (m, U) where:

► $m = (Q, \Sigma, \Delta, q_0, F)$

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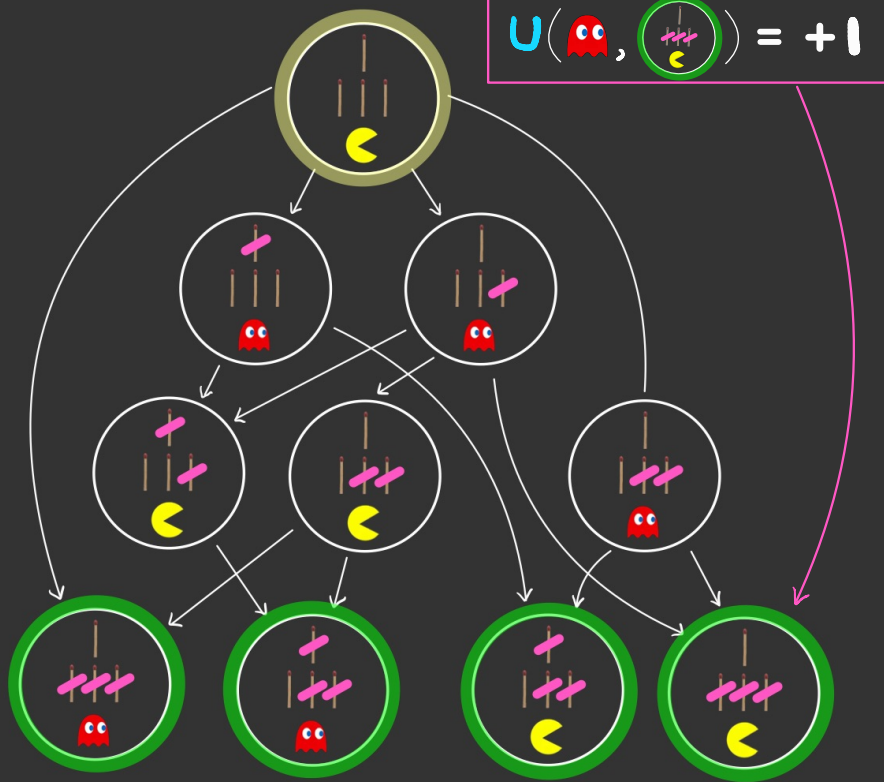
a game with players P
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► $m = (Q, \Sigma, \Delta, q_0, F)$

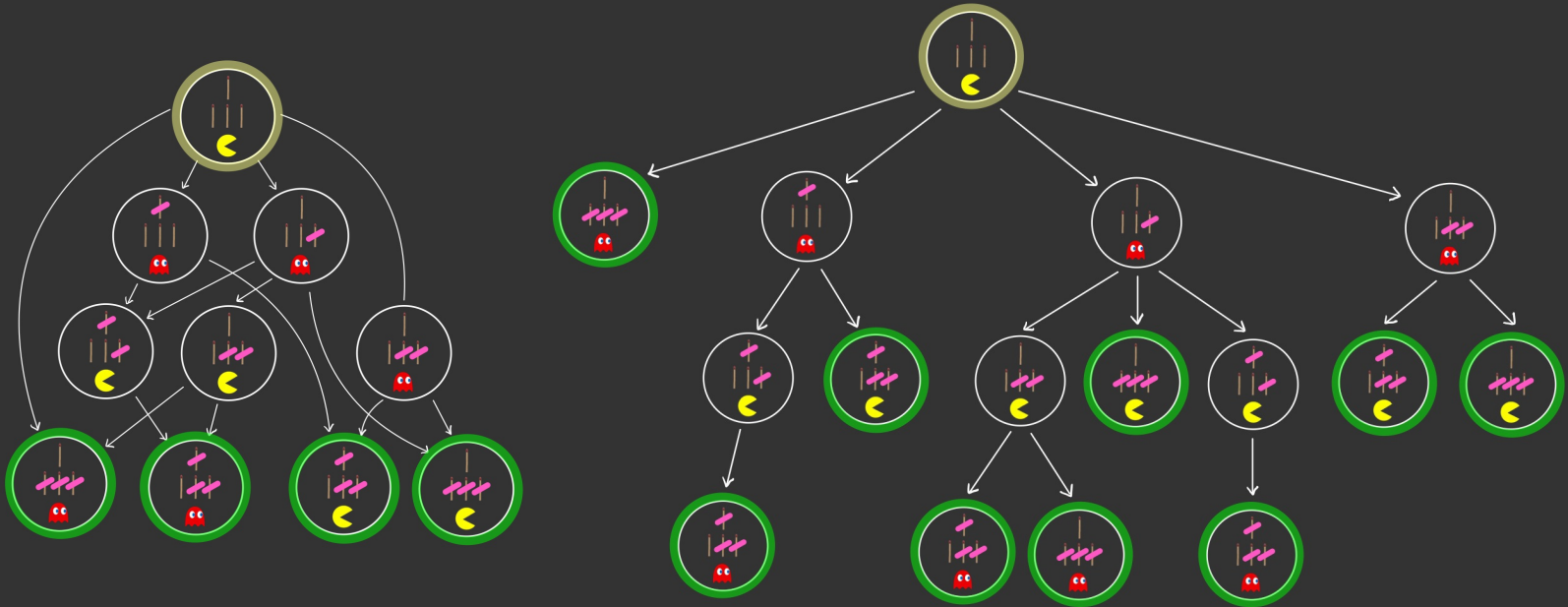
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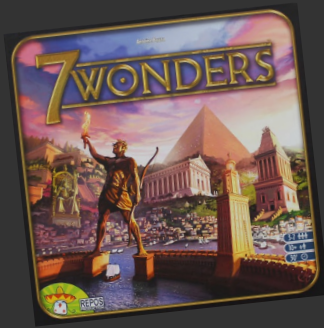
analogous to single-agent search, a game can be unraveled into a search tree



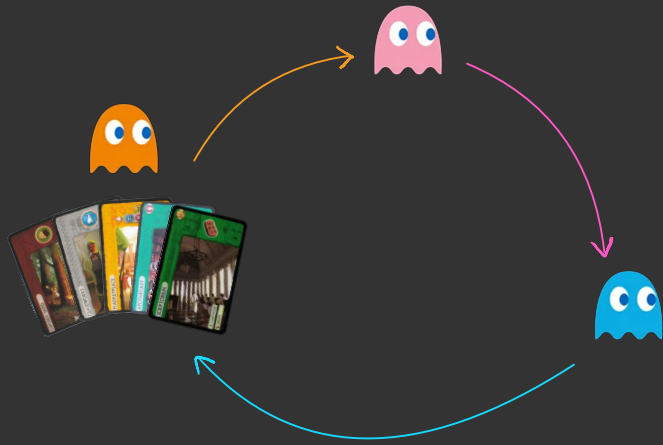
let's see another example



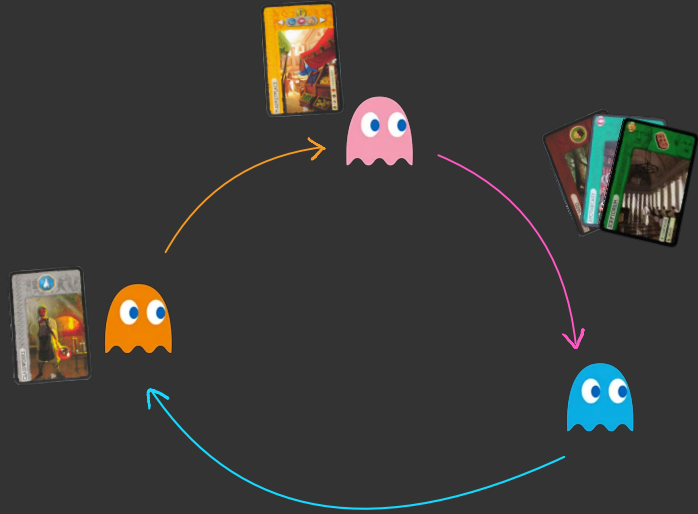
card drafting is a game mechanic
used in many modern games



a set of cards is passed
from player to player



when the player receives the set,
they select one to keep
and pass the rest along



ROYAL FLUSH

This hand contains five cards in sequence, all of the same suit.



STRAIGHT FLUSH

This hand contains five cards in sequence, all of the same suit.



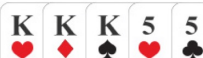
4 OF A KIND

This hand contains all four cards of one rank and any other unmatched card.



FULL HOUSE

This hand contains three matching cards of one rank and two matching cards of another rank.



FLUSH

This hand contains all five cards are of the same suit, but not in sequence.



STRAIGHT

This hand contains five cards of sequential rank in at least two different suits.



3 OF A KIND

This hand contains three cards of the same rank, with two cards not of this rank nor the same as each other.



2 PAIR

This hand contains two cards of the same rank, plus two cards of another rank.



1 PAIR

This hand contains two cards of one rank, plus three cards which are not of this rank nor the same.

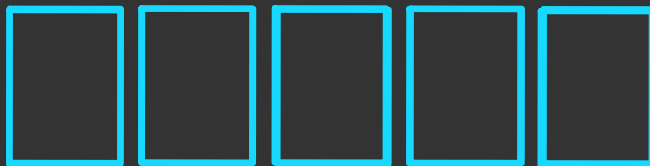


HIGH CARD

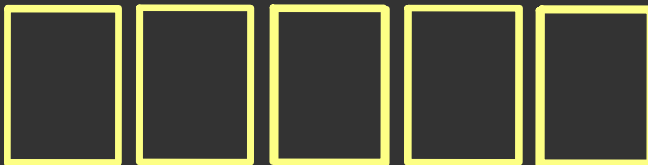
Made of any five cards not meeting any of the above requirements.



your hand



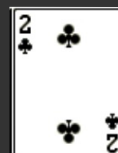
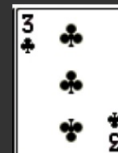
let's draft
poker hands!



my hand



cards



let's consider a two-card version of poker where

STRAIGHT		PAIR		HIGH CARD
 	beats	 	beats	 

draw the search tree for the drafting game where

these are the
players:

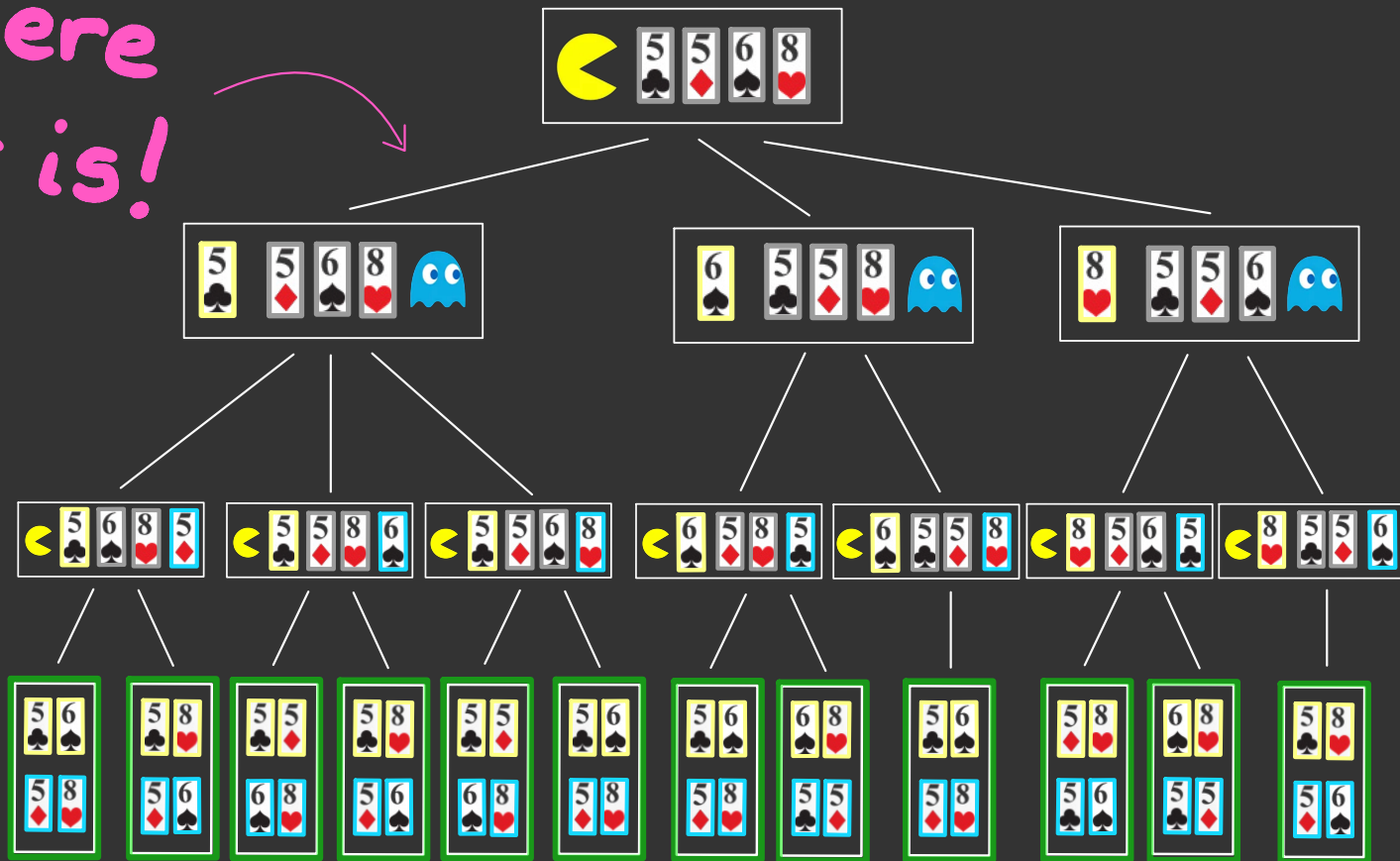
 

these are the
cards:

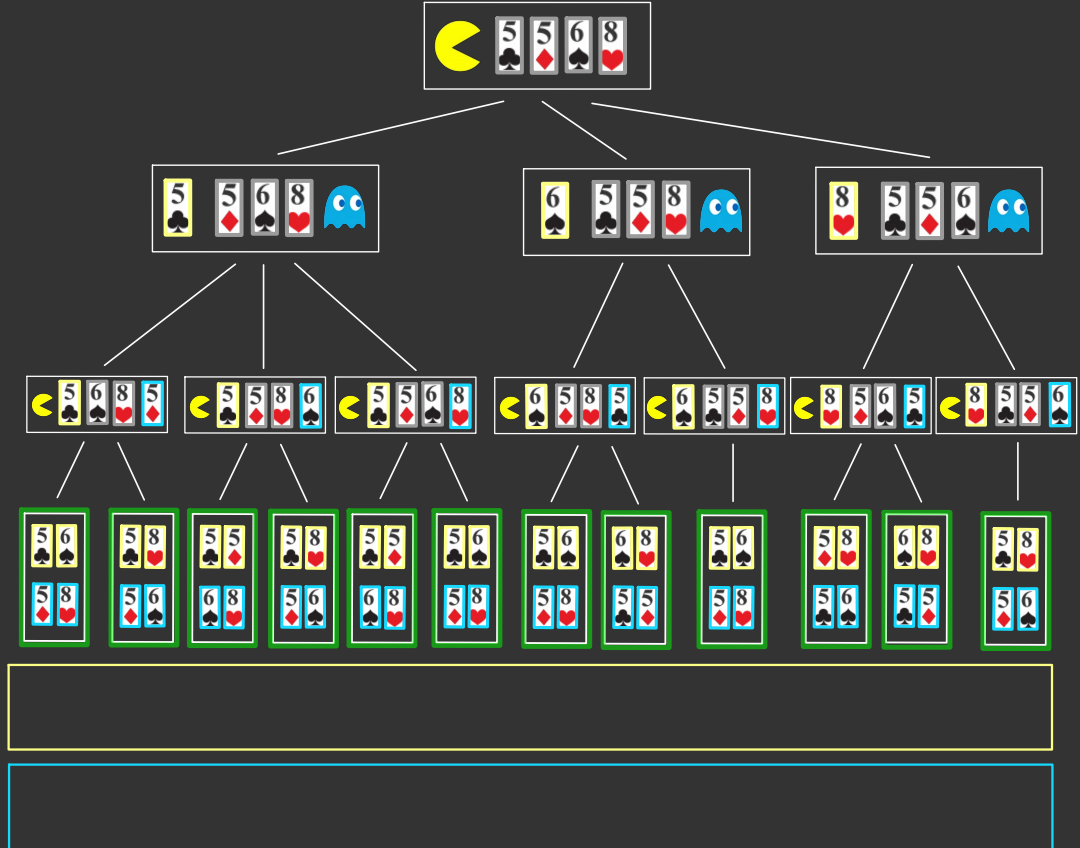
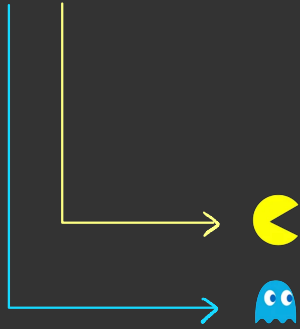
   

goes first

here
it is!



what are the
utilities
of the final
states?



what are the
utilities
of the final
states?

